

1. Administrative & Study Identification

Full Study Title: An Open-label Extension Study to Assess the Efficacy and Safety of Ianalumab With or Without Study Treatment Withdrawal in Adult Participants With Lupus Nephritis Who Have Completed Study Treatment in the CVAY736K12301 Core Study (SIRIUS-LN Extension) **Short Title/Acronym:** SIRIUS-LN Extension **Protocol Number:** CVAY736K12301E1 **NCT Number:** NCT06711887 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Ianalumab (VAY736) 300 mg (Human monoclonal antibody directed against B cells and binding to BAFF receptor) **Phase of Development:** Phase III **Medical Monitor:** Novartis Pharmaceuticals Study Contact

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety after study treatment withdrawal in patients with lupus nephritis (LN) who achieved response (Complete Renal Response [CRR] or Partial Renal Response [PRR]) on double-blind treatment at the end of the SIRIUS-LN core study; AND to evaluate the efficacy and safety of open-label ianalumab 300 mg treatment in patients who either were already receiving ianalumab or did not meet CRR/PRR criteria at the end of the core study. **Secondary Objectives:** To measure sustained remission, renal flares, safety off-study treatment, and long-term tolerability of ianalumab over the 2-year extension period.

2.2 Background & Rationale To address the long-term management of patients with active Lupus Nephritis (LN). While standard therapies and novel biologics like ianalumab can induce remission, it is crucial to determine if patients who achieve clinical responses (CRR or PRR) can safely maintain sustained remission after the withdrawal of the biologic therapy. Additionally, the study evaluates if continued open-label treatment benefits those who did not achieve optimal initial responses.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Open-Label Extension Study) **Control Type:** Single-arm (Treatment withdrawal arm vs. Open-label continuation arm) **Blinding:** Open-label **Randomization:** N/A (Determined by response status at the end of the core study)

3.2 Planned Enrollment & Duration Target **\$N\$:** 315 adult participants **Number of Sites:** Global multicenter (e.g., approximately 25 locations) **Estimated Study Start:** 19-May-2025 **Estimated Primary Completion:** 16-Jul-2032

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Signed informed consent prior to participation in the extension study. 2. Participants must have participated in the SIRIUS-LN core study and must have completed the entire treatment up to Week 144 on double-blind or open-label study treatment.

4.2 Exclusion Criteria 1. Use of prohibited therapies. 2. Pregnant or nursing (lactating) women.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Ianalumab 300 mg administered monthly, OR complete withdrawal of study treatment for eligible responders (maintaining standard of care medications as required). **Route of Administration:** Subcutaneous (SC) injection **Duration of Treatment:** 2 additional years in the extension study.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care (SoC) medications for Lupus Nephritis as required (e.g., mycophenolate mofetil, corticosteroids, tenofovir alafenamide). **Prohibited:** Use of protocol-defined prohibited immunomodulatory therapies and live vaccines.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
:---	:---	:---
:---	:---	Screening/Roll-over Day 0 Informed Consent, verification of Week 140/144 response status (CRR/PRR) from the core study
		Baseline Day 1 Treatment withdrawal initiation (for responders) or open-label ianalumab 300 mg continuation
Interim	Q4W to Q12W	Assessment of sustained remission, renal flares, UPCR, eGFR, safety monitoring, and SoC medication review
End of Study	Year 2	Final efficacy/safety assessments, monitoring for B-cell recovery, and exit interview

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Long-term efficacy (sustained remission and flare rates) and safety profile in participants who undergo treatment withdrawal versus those who continue open-label

ianalumab over the 2-year extension period. **Measurement Method:** Clinical assessment of Complete/Partial Renal Response (CRR/PRR), renal flare tracking, and laboratory monitoring of renal function and B-cell recovery.

7.2 Secondary & Exploratory Endpoints * Time to first renal flare after study treatment withdrawal. * Proportion of patients maintaining CRR or PRR status over time. * Incidence of Adverse Events (AEs), Serious Adverse Events (SAEs), and local injection site reactions.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via scheduled visits, tracking of injection site reactions, and infections. Safety monitoring continues off-treatment until full B cell recovery. **Serious Adverse Events (SAEs):** Routine 24-hour reporting protocols for serious adverse events and severe infections. **Data Monitoring Committee (DMC):** Internal/External safety review mechanisms managed by Novartis Pharmaceuticals.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for efficacy (sustained remission rates) and Safety Set for AE/SAE evaluations among the rolled-over participants. **Sample Size Justification:** Target \$N = 315\$, based on the anticipated eligible completion cohort from the CVAY736K12301 core study. **Handling of Missing Data:** Standard protocol-defined imputation techniques for longitudinal open-label extension data (e.g., LOCF or Mixed Models for Repeated Measures).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to ensure protocol adherence, especially regarding flare detection and SoC adjustments. **Audit Trail:** Robust tracking of eCRF locking, laboratory parameters, and data clarification forms via standard quality audit mechanisms.

11. Study Identifiers & Governance

Primary ID: {{CVAY736K12301E1}} **Registry Number:** {{NCT06711887}}
Sponsor/Lead Organization: {{Novartis Pharmaceuticals}} **Study Title:** {{SIRIUS-LN Extension}} **Official Regulatory Title:** {{An Open-label Extension Study to Assess the Efficacy and Safety of

Ianalumab With or Without Study Treatment Withdrawal in Adult
Participants With Lupus Nephritis Who Have Completed Study
Treatment in the CVAY736K12301 Core Study (SIRIUS-LN Extension))}}

12. Clinical Framework (Lupus Nephritis Specific)

Primary Condition: {{Lupus Nephritis}} **Diagnosis Threshold:** {{Completion of core study Week 144 & evaluation of CRR/PRR status}} **Medical Methodology:** {{Standard of Care (SoC) with or without Open-Label Ianalumab}} **Optimization Target:** {{Sustained Renal Remission / Flare Prevention}}

13. Study Procedures & Methodology

Management Pathway: {{SIRIUS-LN extension management pathway and treatment withdrawal protocol}} **Education Protocol (HCP):** {{Protocol training for flare monitoring and SoC adjustment criteria}} **Education Protocol (Patient):** {{Informed consent, flare symptom awareness, and extension guidelines}} **Quality Audit Mechanism:** {{Clinical monitoring, B-cell recovery tracking, and safety reporting}}

14. Observation & Follow-Up Schedule

Total Duration: {{2 years (Extension Period)}} **Onsite Visit Cadence:** {{Determined by extension protocol (e.g., quarterly follow-ups)}} **Remote Monitoring Frequency:** {{As specified by sponsor monitoring plan}} **Checklist Review Interval:** {{Regular intervals over 2 years for flare and AE assessment}}

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				